

1. Objective

To apply feature scaling, train multiple regression models, compare their performance, and select the best model for house price prediction.

2. Dataset Description

- California Housing Dataset
- 20,640 records
- 8 features
- Target: House Price

3. Data Preprocessing

- Loaded dataset
- Defined features and target
- Split into training and testing data
- Applied Standard Scaler

4. Models Used

- Linear Regression
- Ridge Regression
- Decision Tree Regressor

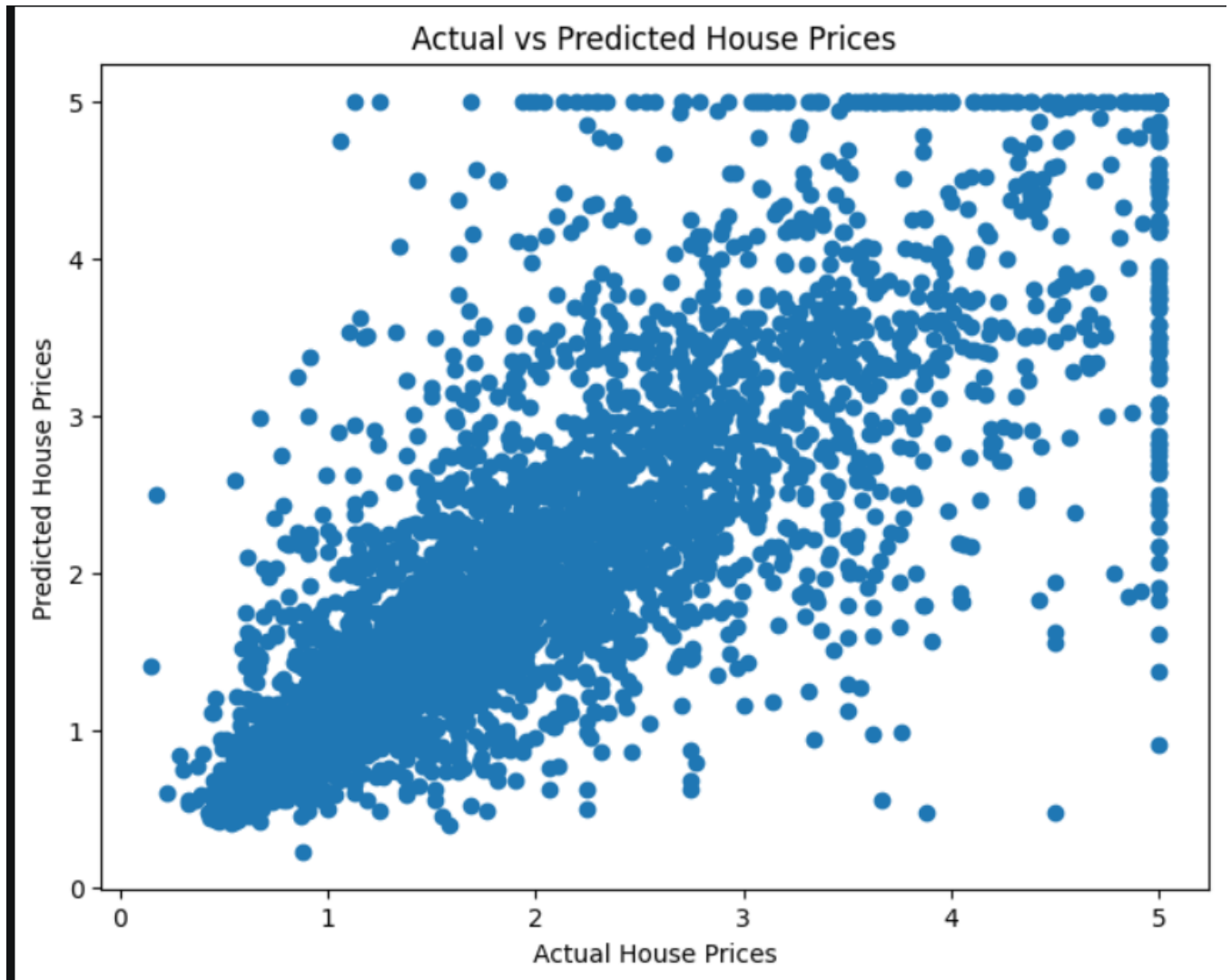
5. Model Evaluation Results

Model	RMSE	R ² Score
Linear Regression	0.745581	0.575788
Ridge Regression	0.745557	0.575816
Decision Tree Regressor	0.702131	0.623791

6. Best Model Selection

The Decision Tree Regressor achieved the lowest RMSE (0.702131) and highest R² Score (0.623791). Therefore, it was selected as the best-performing model.

7. Actual vs Predicted Graph



8. Conclusion

In this project, feature scaling and model optimization techniques were applied to the California Housing Dataset. Three machine learning models were trained and evaluated using RMSE and R^2 Score. After comparing their performance, the Decision Tree Regressor was identified as the best-performing model. The project demonstrates the importance of model comparison in improving predictive accuracy.